

Embedded Android Gets Popular!

This article explains why Android is the preferred platform for modern embedded solutions. It also covers the extra market edge that you get with Android.



The Android mobile platform is being adopted by a broad range of embedded devices that span multiple industries and segments. Android-enabled custom solutions, such as the Kindle Fire by Amazon, have proven to be game-changing in the industry. According to Amazon, the Kindle Fire captured 22 per cent of the US tablet market, and increased the company's e-books sales by 175 per cent in 2011.

The rise of modern embedded systems

As we know, embedded systems control many devices that are in common use today. They range from portable devices such as digital watches and MP3 players, to large stationary installations like ATM and vending machines. However, embedded systems have changed dramatically in recent years. Today, they are largely media-rich, connected and highly integrated. Many include graphical user interfaces with high-resolution 2D and 3D graphics. Additionally, nearly all embedded systems include IP networking stacks, and link connectivity via a combination of wired and wireless network interfaces. The core feature sets often rely

upon connectivity. And last of all, for reasons such as power efficiency, size and performance, chipsets for embedded systems are designed to be highly integrated.

This drastic change in the characteristics of modern embedded systems has given rise to advanced functionality and user experience needs, and Android helps address these needs.

Android's characteristics

Here are some of the key characteristics of the Android platform that make it suitable for modern embedded systems.

Dalvik Virtual Machine

Android's Dalvik Virtual Machine is specifically designed to support a diverse set of devices, where the applications must be sand-boxed for security, performance and reliability. Also, it works very well with limited processor speeds and RAM.

Hardware platforms support

Android supports a variety of hardware platforms. Apart from ARM-based Android development phones, it also supports hardware platforms for prototyping and